Moments of Inertia

COMMON SECTIONS AND BODIES

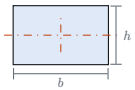
A statics · second moment of area
B dynamics · second moment of mass

$I_x = \int_A y^2\,\mathrm{d}A$ | $I_y = \int_A x^2\,\mathrm{d}A$ | $J_O = I_x + I_y$ | $I = \bar{I} + Ad^2$ | $k = \sqrt{I/A}$

A — Area moments

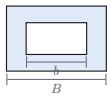
$I = \int_A y^2\,\mathrm{d}A$ · section property, mm⁴ · bar = centroidal axis, plain = axis shown

Rectangle $b \times h$



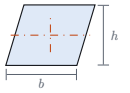
$A = bh$
$I_x = (bh^3)/12$
$I_y = (hb^3)/12$
$J_O = (bh(b^2+h^2))/12$

Hollow rectangle



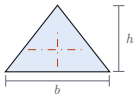
$A = BH - bh$
$I_x = (BH^3 - bh^3)/12$
$I_y = (HB^3 - hb^3)/12$

Parallelogram $b \times h$



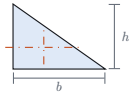
$A = bh$
$\bar{I}_x = (bh^3)/12$
$I_{\text{base}} = (bh^3)/3$

Isosceles triangle



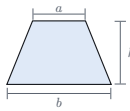
$A = bh/2$
$I_{\text{base}} = (bh^3)/12$
$\bar{I}_x = (bh^3)/36$
$\bar{I}_y = (hb^3)/48$

Right triangle



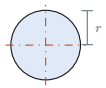
$A = bh/2$
$\bar{I}_x = (bh^3)/36$
$\bar{I}_y = (hb^3)/36$
$\bar{I}_{\text{xy}} = -(b^2h^2)/72$

Trapezoid



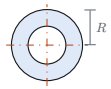
$A = ((a+b)h)/2$
$\bar{I}_x = (h^3(a^2+4ab+b^2))/(36(a+b))$
$\bar{y} = (h(2a+b))/(3(a+b))$

Circle r



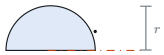
$A = \pi r^2$
$I_x = I_y = (\pi r^4)/4$
$J_O = (\pi r^4)/2$
$I_x = (\pi d^4)/64$

Annulus R, r



$A = \pi(R^2 - r^2)$
$I_x = I_y = (\pi(R^4 - r^4))/4$
$J_O = (\pi(R^4 - r^4))/2$

Semicircle r



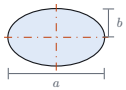
$A = (\pi r^2)/2$
$I_{\text{base}} = (\pi r^4)/8$
$\bar{I}_x \approx 0.1098 r^4$
$\bar{y} = (4r)/(3\pi)$

Quarter circle r



$A = (\pi r^2)/4$
$I_{\text{edge}} = (\pi r^4)/16$
$\bar{I} \approx 0.0549 r^4$
$\bar{x} = \bar{y} = (4r)/(3\pi)$

Ellipse a, b



$A = \pi ab$
$I_x = (\pi ab^3)/4$
$I_y = (\pi a^3 b)/4$

Thin-walled tube

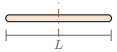


$A \approx 2\pi Rt$
$I \approx \pi R^3 t$
$J_O \approx 2\pi R^3 t$
$t \ll R$

B — Mass moments

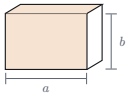
$I = \int r^2\,\mathrm{d}m$ · rigid-body property, kg · m² · principal axis named on each line

Thin rod, length L



$\bar{I} = (mL^2)/12$
$I_{\text{end}} = (mL^2)/3$

Rectangular plate



$I_z = (m(a^2+b^2))/12$
$I_{\text{edge}} = (ma^2)/3$

Disk / ring



disk $I_z = \frac{1}{2}mR^2$
disk $I_{\text{diam}} = \frac{1}{4}mR^2$
ring $I_z = mR^2$

Sphere



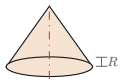
solid $I = \frac{2}{5}mR^2$
thin shell $I = \frac{2}{3}mR^2$
any diameter

Cylinder, length L



$I_{\text{axis}} = \frac{1}{2}mR^2$
$I_{\text{centroid}} = \frac{1}{4}mR^2 + (mL^2)/12$
thin shell $I_{\text{axis}} = mR^2$

Solid cone, height h



$I_{\text{axis}} = \frac{3}{10}mR^2$
$I_{\text{centroid}} = \frac{3}{20}mR^2 + (3mh^2)/80$

Composite bodies $I_x = \sum_i \left(\bar{I}_i + A_i d_i^2\right)$ T-section, flange 100 × 20 on web 20 × 80 (total depth 100 mm): $A = 3600$ mm², $\bar{y} = 67.8$ mm from the base, giving $I_x \approx 3.14 \times 10^6$ mm⁴. d = centroid-to-axis distance $d = 0$ on a centroidal axis